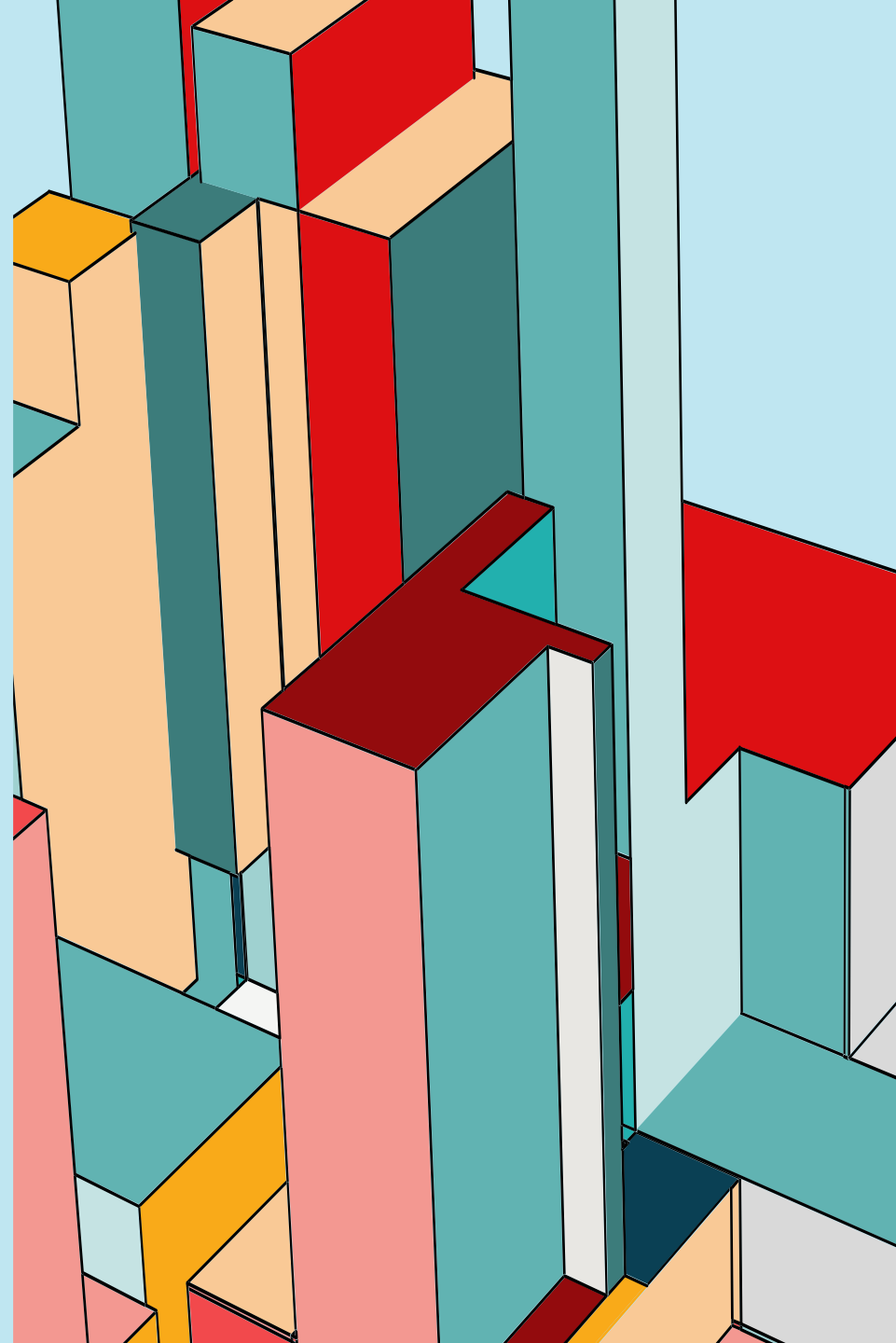


SOFTWARE ARCHITECTURE · arc42

Autoware VisionPilot on AUTOSAR Adaptive

One ROS 2 process decomposed into four Adaptive Applications on CAPI –
as built, as run

Bench: WSL2 / Ubuntu 24.04 x86-64 · CAPI v1.0.0 (AP R20-11)
SOME/IP · shared-memory frame ring



HOW TO READ THIS DECK

arc42 v8 – twelve sections, mapped to the slides that cover them

§1 Introduction & Goals	3	§7 Deployment View	16 – 17
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§3 Context & Scope	5	§9 Architecture Decisions	21
§4 Solution Strategy	6	§10 Quality Requirements	22 – 23
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§6 Runtime View	13 – 15	§12 Glossary	26

NOT COVERED

Localization – no map, no pose
Sensor fusion beyond one camera
Diagnostics / UCM / crypto
Real vehicle actuation

STATUS OF THIS ARCHITECTURE

This is an **as-built** description. The model validates against AUTOSAR_00049.xsd, generates with ara-gen, builds against an installed CAPI SDK, and has been booted: eight processes started by Execution Management, service discovery over SOME/IP, a shared-memory frame ring, and a closed control loop at 20 Hz on a recorded clip.

What is NOT yet done is named explicitly in §11 rather than left for the reader to discover.

§1 INTRODUCTION & GOALS

Why move a working ROS 2 pilot onto an Adaptive Platform at all

Deterministic lifecycle

Execution Management starts and stops processes against declared function-group states. Stopping one component and keeping the others becomes a line in a manifest instead of a supervisor script.

exec · sm

A reviewable interface

Every boundary is an ARXML ServiceInterface with a fixed SOME/IP service ID, event ID and port. The wire format is a versioned artifact a reviewer can read – not a struct that happens to compile.

com · isoft/nsomeip

Failure is a modelled state

Losing the planner is not a crash to be caught – it is a transition to AvPilotFG.Degraded, after which the remaining components keep running and Control safe-stops on its own staleness check.

exec · sm

QUALITY GOALS (§1.2)

GOAL	HOW THIS ARCHITECTURE SERVES IT	EVIDENCE
Safety	A trajectory older than 200 ms must not be applied. Control detects this itself, without relying on the platform.	verified on the bench
Analysability	Every inter-process datum is a named type in the model with a known size.	12 ARXML files, XSD-valid
Modifiability	Stopping or adding a component is a manifest change, not a code change.	8 processes, one FG table
Performance	The perception chain must not be starved by the component it protects.	SCHED_RR, control 22 > perception 12

Goals are stated so they can be falsified – each names the evidence that would settle it.

Source: [@f9fb997](https://github.com/autowarefoundation/vision_pilot) (Apache-2.0) · [@112916a](https://github.com/AUTOSAR/capi)

§2 CONSTRAINTS

Fixed before any design decision was taken

TECHNICAL

CAPI v1.0.0 – AP R20-11

Only binding in the tree:

isoft/nsomeip (SOME/IP)

No DDS binding exists

isoft/npc = local IPC,
not a standardised binding

PLATFORM (BENCH)

WSL2 · Ubuntu 24.04 · x86-64

GCC 13.3 · CMake 3.28

Debug SDK build

No GPU – ONNX on CPU

No real-time kernel

Shared host with Windows

ORGANISATIONAL / LEGAL

CAPI: INFORMATION ONLY

Commercial use needs an
AUTOSAR partnership

VisionPilot: Apache-2.0

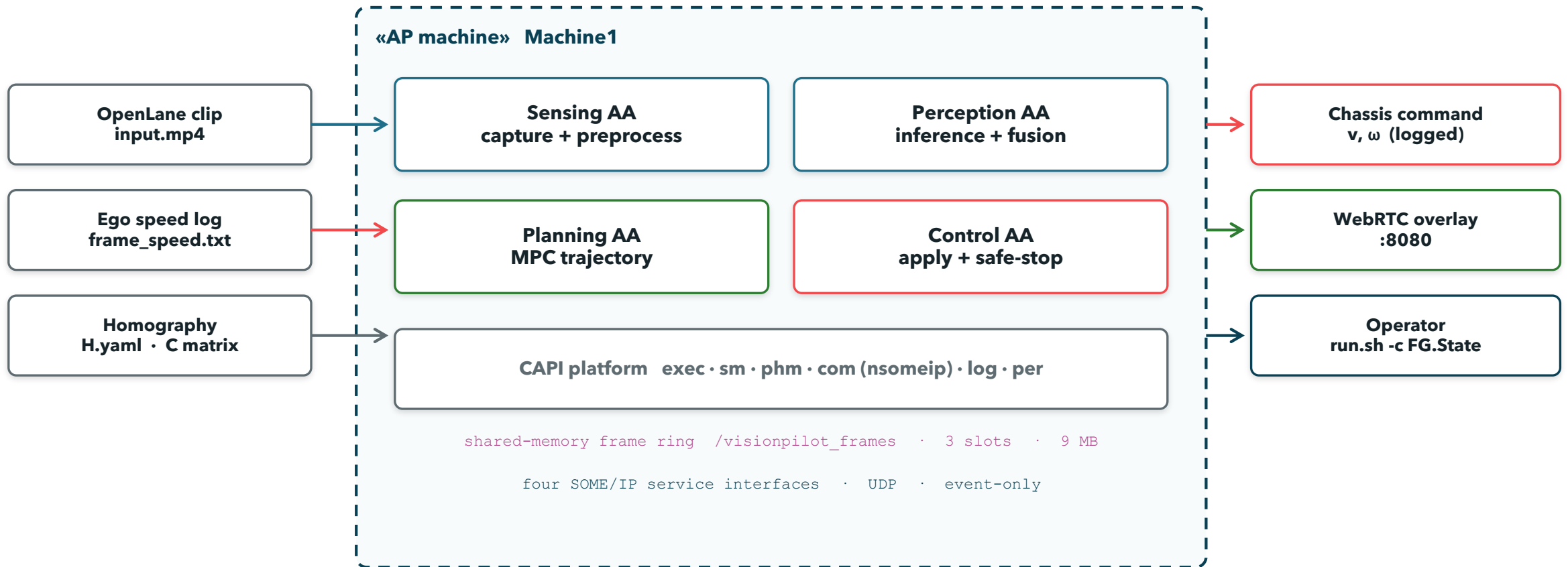
Both consumed as pinned
submodules, never vendored

CONSEQUENCE SOME/IP is the only standardised binding CAPI ships, so every modelled interface must fit a UDP datagram. A 1024×512×3 camera frame is 1.5 MB and cannot. That single constraint is why Sensing and Perception share memory instead of a service – and why the interface between them carries a **34-byte descriptor** rather than an image.

BENCH ≠ TARGET Everything in this deck was run on a WSL2 development bench against a recorded clip. No number here is a target-hardware measurement, and the deck does not claim one. Timing, scheduling and thermal behaviour must be re-established on the real ECU before any of it is quoted.

§3 CONTEXT & SCOPE

System boundary – what runs inside the AP machine and what crosses it



OUT OF SCOPE Real actuation (Control logs what it would write) · localization · LiDAR and radar · UCM, diagnostics, crypto, network management – all present in CAPI, none exercised here.

§4 SOLUTION STRATEGY

Four decisions that shape everything downstream

Split along the reference AD blocks

Where Sensing, Perception, Planning and Control are separate Adaptive Applications, matching the decomposition every AD stack uses. Sensing is the sensor abstraction – camera today, radar or LiDAR later – not a part of Perception.

Roles 4 AAs · 4 executables · 8 processes

Frames by shared memory, not by service

Where A 1.5 MB frame cannot cross SOME/IP. Sensing writes into a seqlock ring in /dev/shm and publishes a 34-byte descriptor naming the slot. The interface is fully modelled; only the buffer transport is not.

Roles FrameDescriptor 34 B · 3 slots · 9 MB

Model the states, do not script them

Where One function group, AvPilotFG, with five states. Which process runs in which state is a table in the execution manifest. SetState(Degraded) stops Planning and leaves the other three running – no C++ implements that.

Roles Off · Init · Driving · Degraded · Safe

Bound every type on the wire

Where Each event is one datagram: DrivingState 47 B, Trajectory 97 B, VehicleSpeed 17 B. Trajectory carries the whole 20-step steering horizon so Control can keep steering after Planning stops.

Roles ara::core::Array<float,20> · no fragmentation

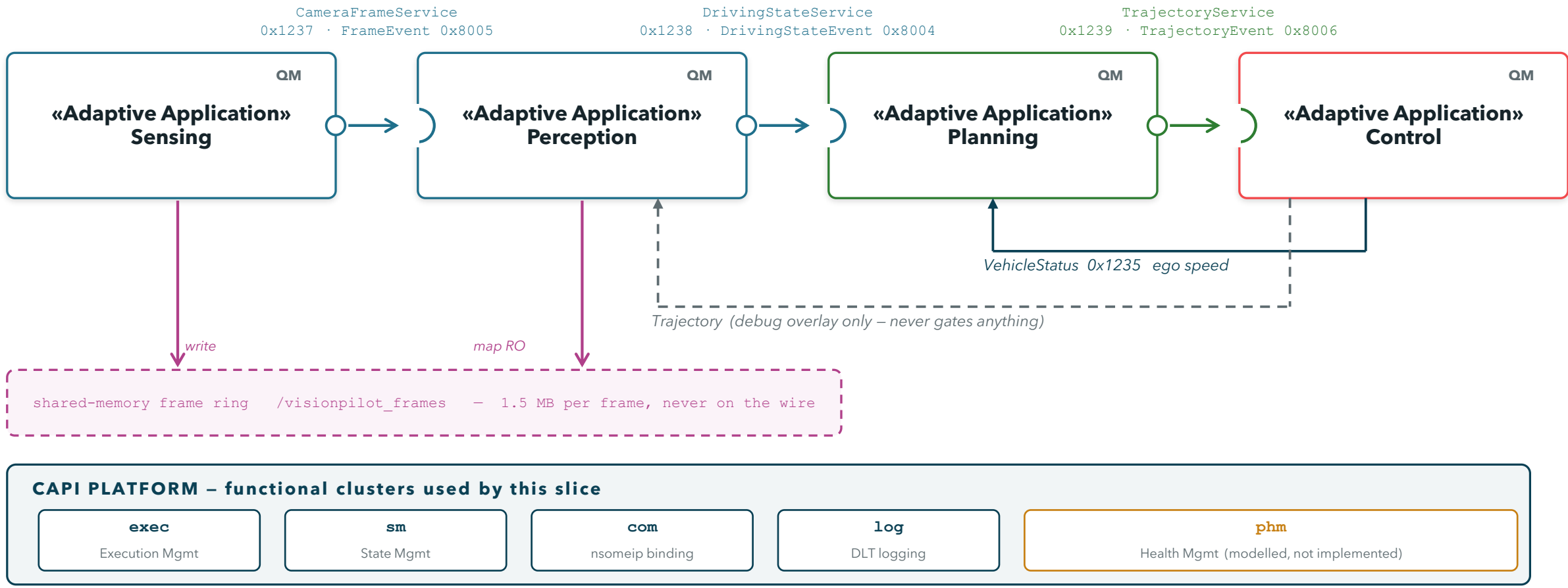
TAKEN TOGETHER Four processes with a declared state table, bounded events on the wire, bulk data out of band, and a consumer that judges the freshness of its input independently. The degrade is then a state transition rather than an exception handler – which is the one property of this design that ROS 2 cannot express.

“Where” states the decision · “Roles” states the mechanism – the rejected alternatives are recorded in §9.

Source: this project's AUTOSAR model — av-stack/model/.arxml, validated against AUTOSAR_00049.xsd and generated with ara-gen*

§5.1 BUILDING BLOCK VIEW – LEVEL 1

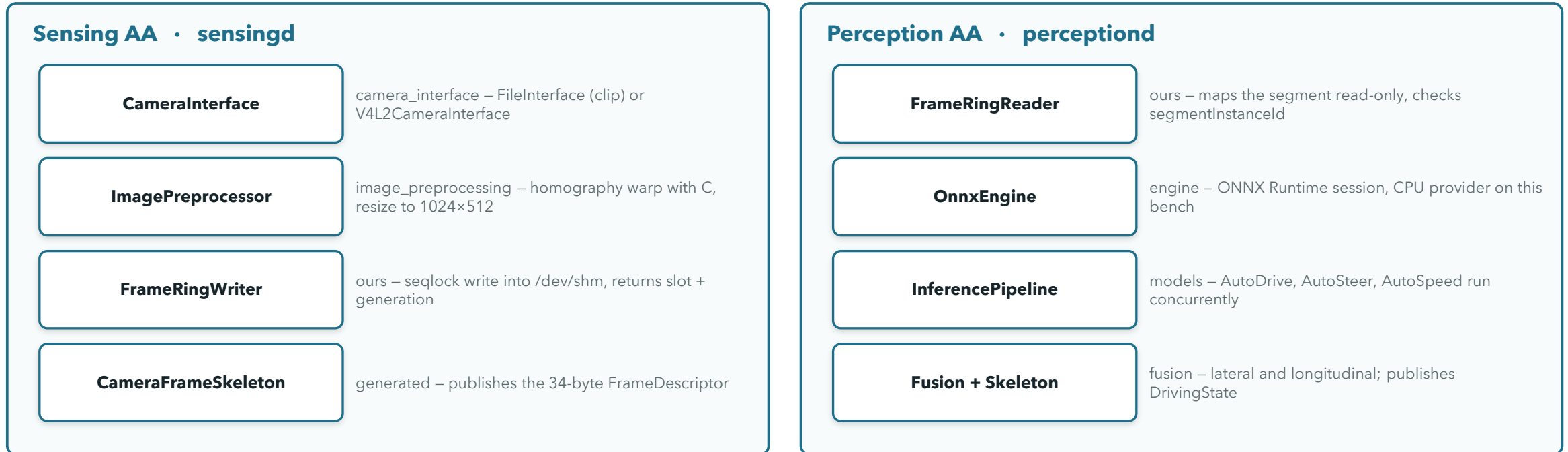
Four Adaptive Applications over the CAPI platform · provided ● / required ⊃ ports



Every arrow above is an event-only ServiceInterface bound to SOME/IP UDP by deployment – no methods, no fields.
Source: this project's AUTOSAR model — av-stack/model/*.arxml, validated against AUTOSAR_00049.xsd and generated with ara-gen

§5.2 BUILDING BLOCK VIEW – LEVEL 2 · SENSING & PERCEPTION

White box – which upstream VisionPilot module ended up in which process

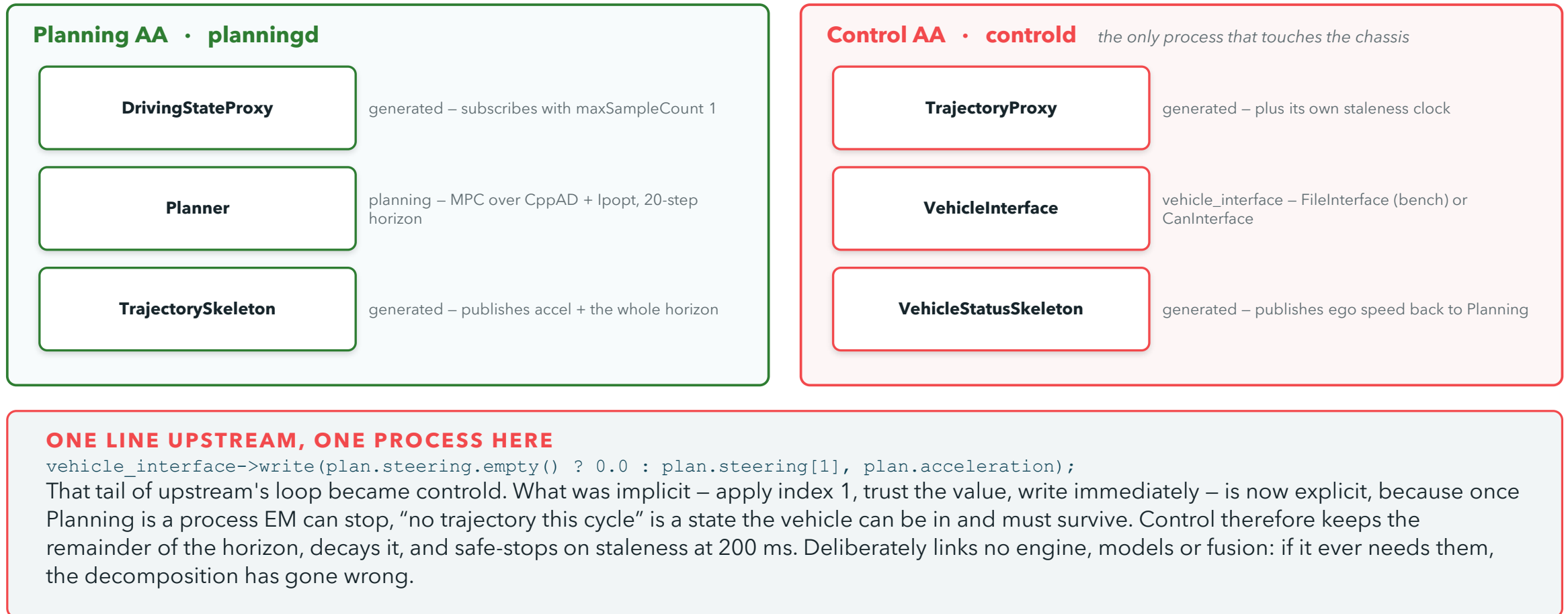


WHY PREPROCESSING MOVED INTO SENSING Upstream runs capture → preprocess → inference in one thread, so wall time is pre + parallel. With preprocess in Sensing the two stages pipeline across processes and steady-state throughput becomes max(pre, parallel) instead of their sum.

It also moved a hidden coupling into the open: the warp is mounting-specific, so it belongs to the sensor abstraction – but Perception still needs the **original capture dimensions** to project lane output back to world. See §11.

§5.2 BUILDING BLOCK VIEW – LEVEL 2 · PLANNING & CONTROL

The two processes that decide and the one that acts



§5.3 INTERFACE CONTRACT

Four ServiceInterfaces – all event-only, all SOME/IP UDP, all one datagram

SERVICE INTERFACE	EVENT	SERVICE ID	EVENT ID	PAYLOAD	PROVIDER → CONSUMER
CameraFrameService	FrameEvent	0x1237	0x8005	34 B	sensingd → perceptiond
DrivingStateService	DrivingStateEvent	0x1238	0x8004	47 B	perceptiond → planningd
TrajectoryService	TrajectoryEvent	0x1239	0x8006	97 B	planningd → controld (+ perceptiond, debug)
VehicleStatusService	SpeedEvent	0x1235	0x8002	17 B	controld → planningd

WHY EVENTS AND NOT FIELDS

Every one of these is a stream where only the latest sample matters.
A Field would add get/set semantics and an initial value nobody needs.

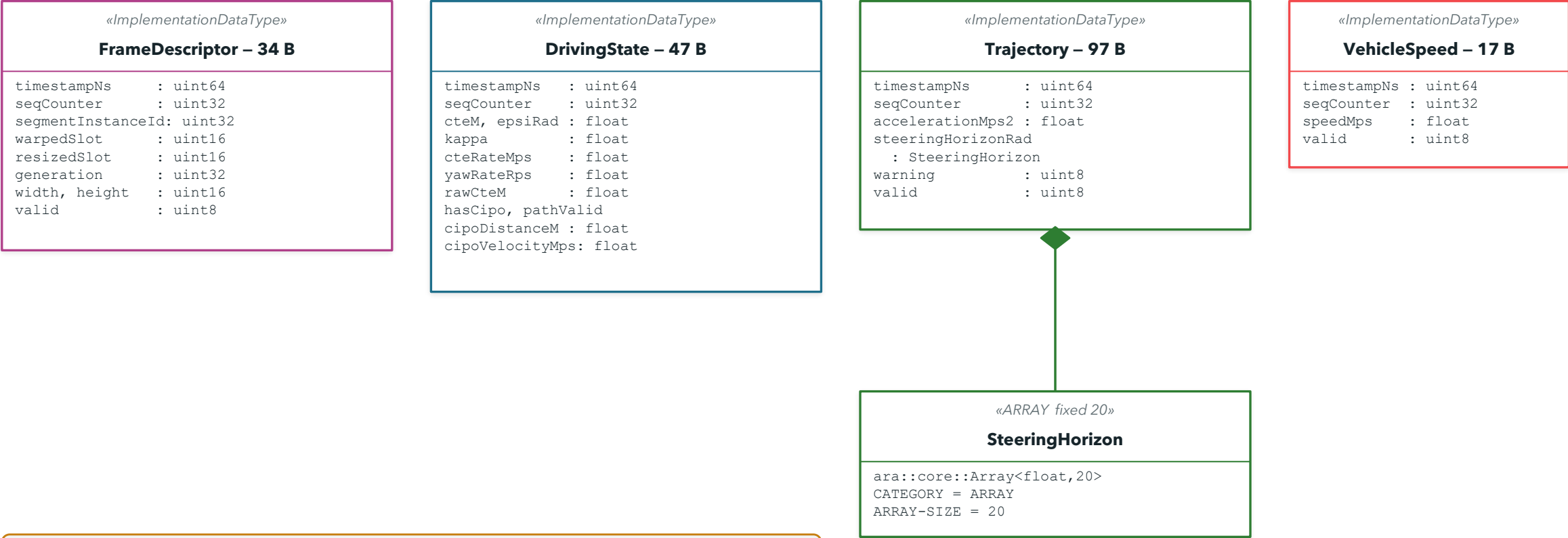
WHY EVENT-ID < 0X8000 IS WRONG

SOME/IP reserves the top bit for notifications. ara-gen accepts a smaller id and the event then never arrives – a silent discovery fault.

DELIBERATELY UNALLOCATED 0x1234 and 0x1236 are left free. CAPI's own helloworld-cm sample uses 0x1234; colliding with it makes two unrelated services indistinguishable on the wire during bring-up.

§5.4 DATA TYPES

Every inter-process datum is a named *ImplementationDataType* with a known size

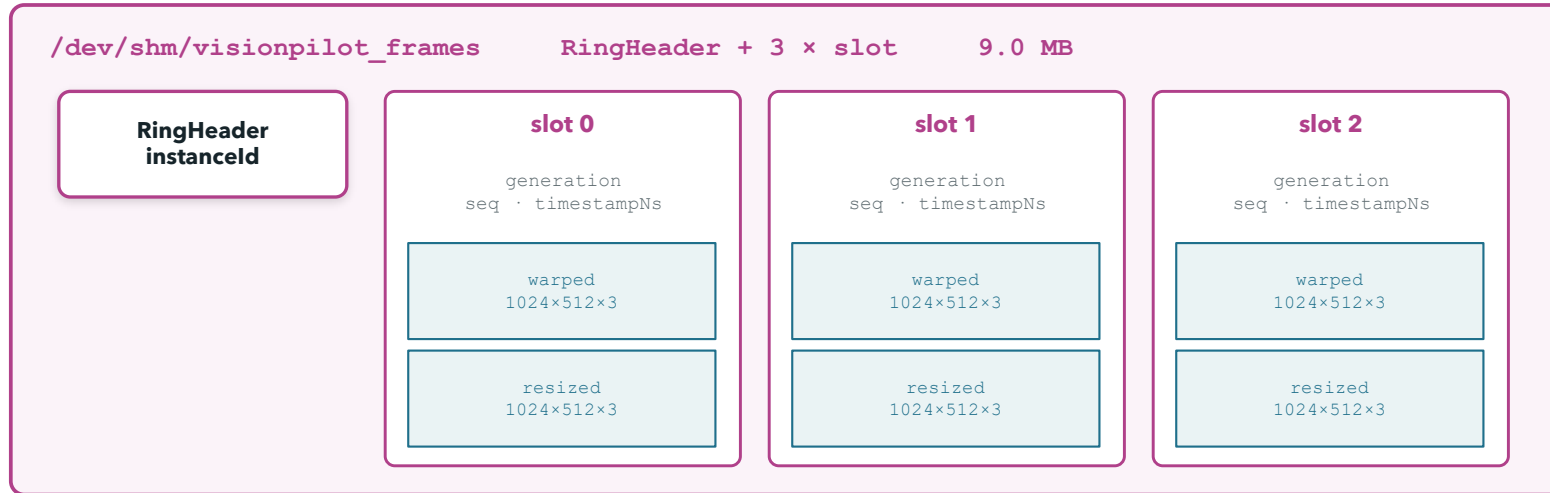


ARXML ORDER MATTERS ARRAY-SIZE must precede NAMESPACES inside STD-CPP-IMPLEMENTATION-DATA-TYPE. The XSD error lists only what may still follow, so it reads as “this element is not allowed” when the element is merely misplaced.

The whole 20-step horizon is sent, not one value – that is what lets Control keep steering after Planning is stopped.
Source: this project's AUTOSAR model — av-stack/model/*.arxml, validated against AUTOSAR_00049.xsd and generated with ara-gen

§5.5 THE SHARED-MEMORY FRAME RING

The one thing that is not on the wire – and how it stays consistent without a lock



SEQLOCK, NOT A MUTEX

1. slot = seq % 3
 2. generation += 1 (odd)
 3. memcpy warped, resized
 4. generation += 1 (even)
 5. publish descriptor
- Reader re-checks generation;
odd or changed ⇒ discard.

Why 3 slots

Where Minimum that keeps the writer off the slot a reader is holding while a third is being filled. Two would make a lapped read the common case, not the rare one.

Roles `kSlots = 3`

Why an instance id

Where A restarted sensingd recreates a segment that looks identical to the old one. The id changes on every `create()`, so a stale reader detects it instead of reading a recycled buffer.

Roles `segmentInstanceId` in every descriptor

What it costs

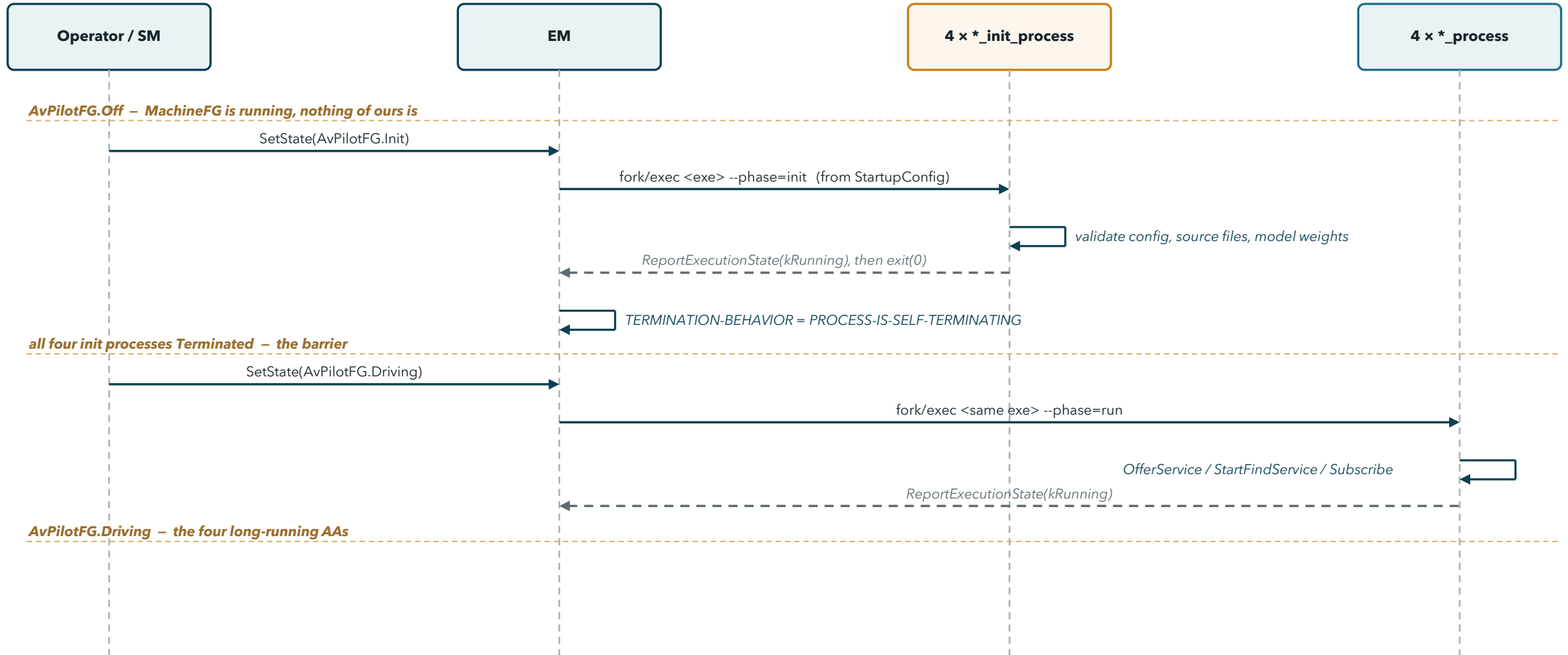
Where This transport is outside the AUTOSAR model. The interface is fully modelled; the buffer is not. On a machine with a zero-copy binding this would be `ara::com`'s job.

Roles `isoft/npc` exists but is not standardised

OBSERVED `sensingd: frame ring created ... instance=441054940` · `perceptiond: found CameraFrameService → frame ring mapped ... instance=441054940` – the same id in both processes is the evidence that discovery and the segment agree.

§6.1 RUNTIME VIEW – TWO-PHASE STARTUP

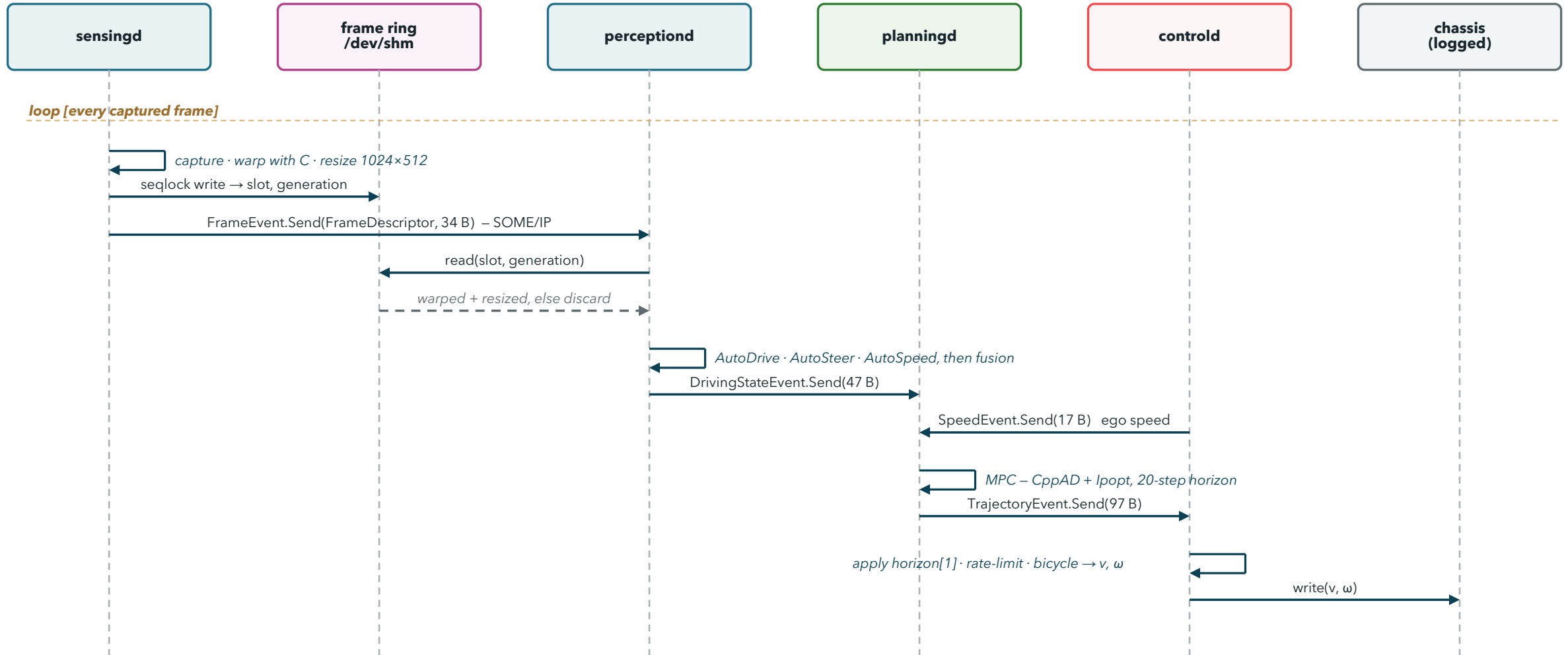
Eight processes over four executables · told apart by the process argument in the execution manifest



Nothing enforces the ordering: Off → Driving is legal and skips validation entirely. The barrier is the function-group state, because an *ExecutionDependency* cannot span two function-group states – so today the operator moves it, not State Management.

§6.2 RUNTIME VIEW – NOMINAL CYCLE

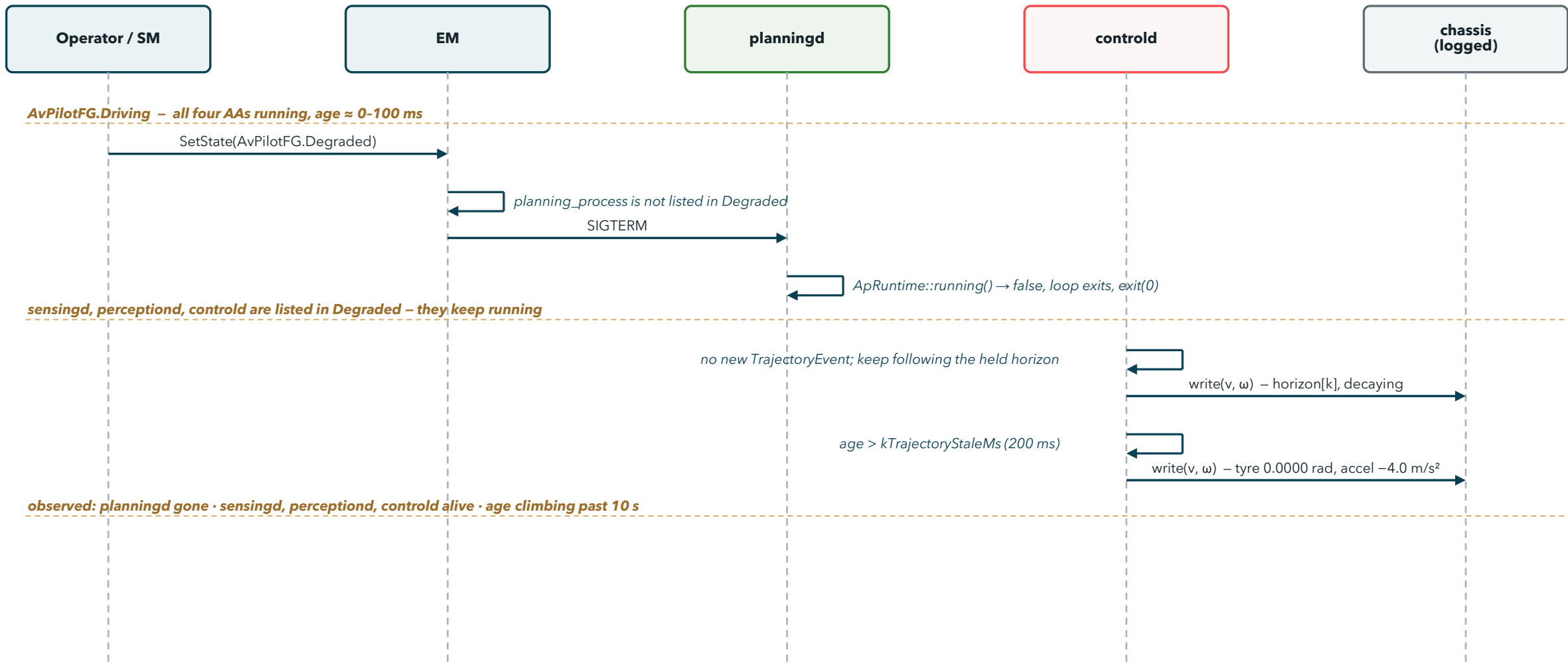
One frame through four processes in AvPilotFG.Driving · Perception ~10 Hz, Control 20 Hz



timestampNs is set at CAPTURE and carried the whole way, so Control's staleness check measures end-to-end age – not the age of the last hop.

§6.3 RUNTIME VIEW – LOSS OF PLANNING → DEGRADE

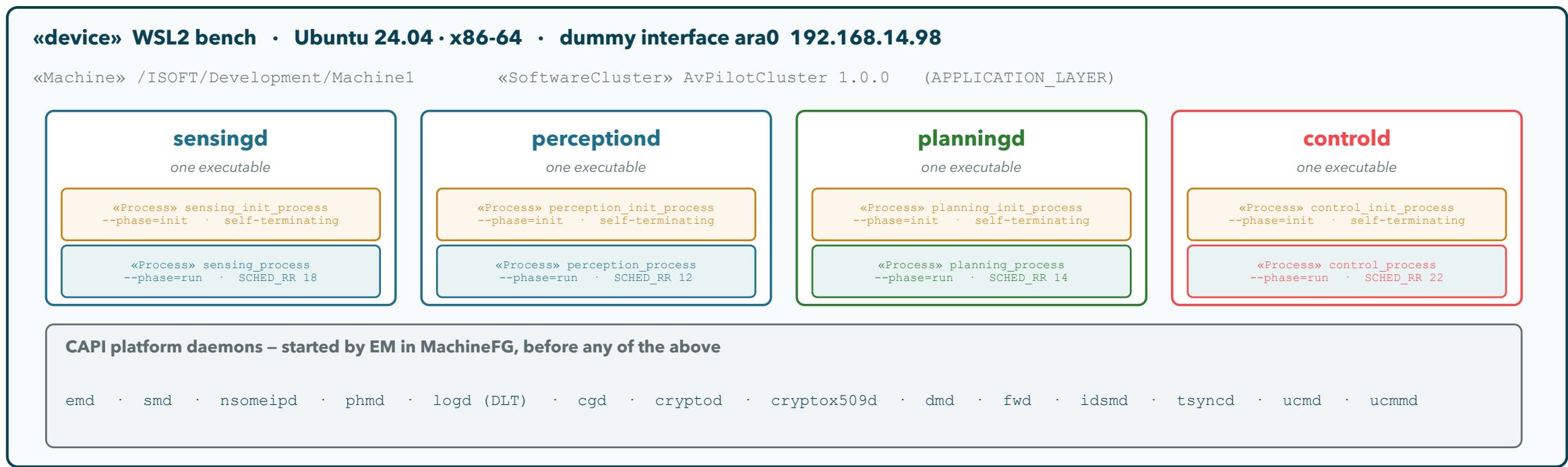
The scenario the architecture exists to express · observed on the bench, not modelled on paper



No C++ implements this transition. It is the eight-row state table in execution_manifest.arxml – and Control's own staleness clock, which does not wait for the platform to tell it anything.

§7.1 DEPLOYMENT VIEW

One machine · one software cluster · eight processes over four executables



PRIORITY ORDER, AND WHY

controld 22 · sensingd 18 · planningd 14 ·
perceptiond 12

Control owns the actuator and is the last gate;
it must not be starved by the chain it protects.

BENCH-ONLY NETWORK DETAIL

ara0 is a dummy interface carrying the SDK's own
default address. WSL2's mirrored adapters refuse raw
sockets, which takes tsyncd and the machine down.

§7.2 THE FUNCTION GROUP TABLE

AvPilotFG – the whole degradation story, declared once in the execution manifest

PROCESS	Off	Init	Driving	Degraded	Safe
sensing_init_process		●			
perception_init_process		●			
planning_init_process		●			
control_init_process		●			
sensing_process			●	●	
perception_process			●	●	
planning_process			●		
control_process			●	●	●

READ THE ASYMMETRY

Driving → Degraded removes planning_process only.
Control is listed in Driving, Degraded AND Safe – so it is the last thing still running.

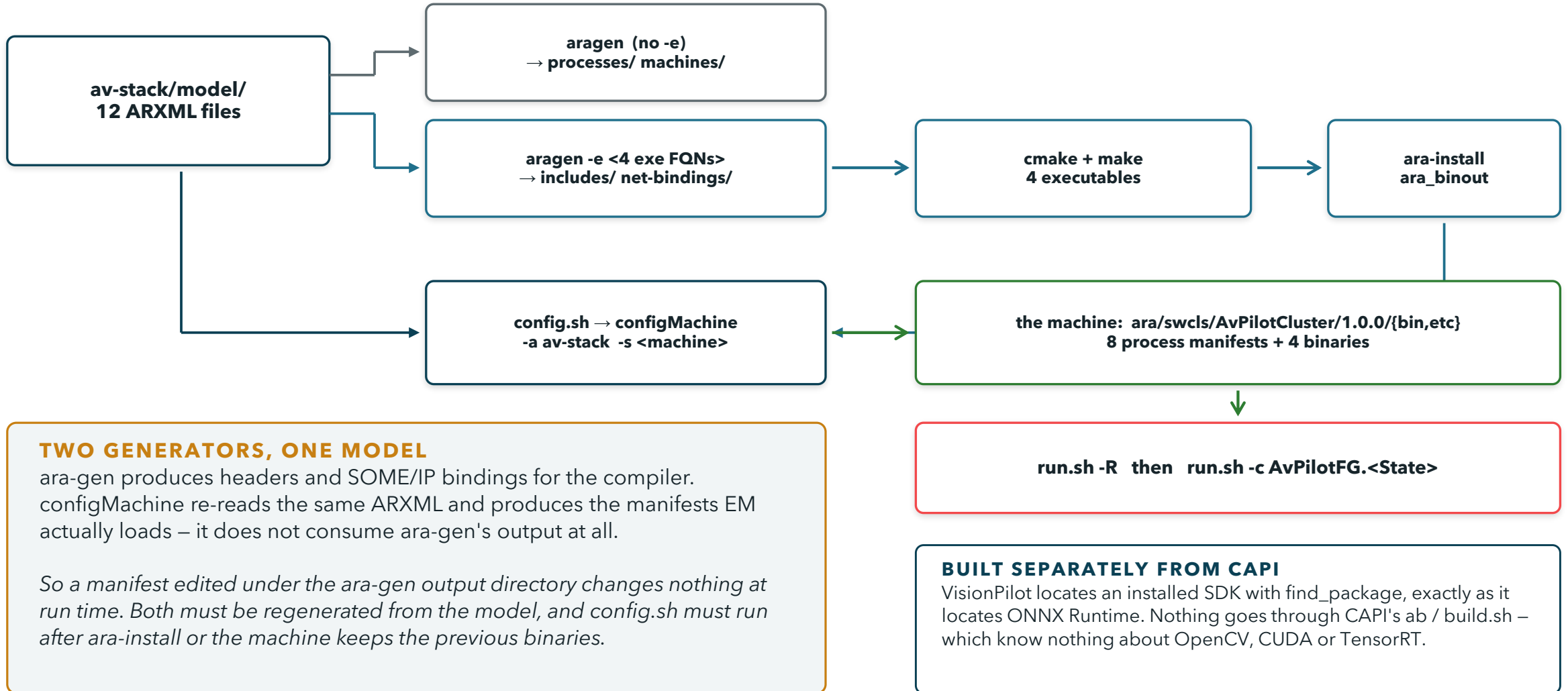
INIT IS ONE-SHOT

The four *_init processes are self-terminating. They validate and exit; nothing they build in memory survives into run.

THIS TABLE IS THE ARCHITECTURE Adding a degraded mode, or moving a component between them, is an edit here plus a regenerate – no application code changes and no supervisor learns a new rule.

§8.1 CROSSCUTTING – FROM MODEL TO RUNNING MACHINE

Two generators read the same ARXML for two different purposes – a trap worth naming



§8.2 CROSSCUTTING – SUPERVISION & SAFETY CONCEPT

One detection path is implemented and observed; the other is designed and not yet built

Staleness check – BUILT

Where Control timestamps every trajectory from CAPTURE and compares age against 200 ms on a CLOCK_MONOTONIC clock. Independent of the platform: it needs nobody to tell it anything.
Roles observed: age 0→51→104 ms nominal, >10 s degraded

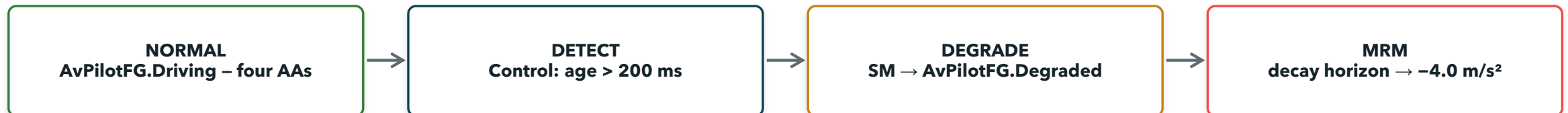
Alive supervision – NOT BUILT

Where PHM would count kCycleStart indications per reference cycle and catch a dead or runaway producer. The checkpoint enum exists in ApRuntime; the supervised entity is not modelled.
Roles ApRuntime("") – checkpoints are dropped

Deadline supervision – NOT BUILT

Where Elapsed kFrameReceived → kPublished would catch the alive-but-late case that liveness cannot see – the throttling failure. Designed, checkpointed in code, unmodelled.
Roles no PHM-SUPERVISED-ENTITY-INTERFACE yet

REACTION LADDER



WHAT THIS CONCEPT IS AND IS NOT

Today there is **one** detection path, in the consumer, and it works. The two PHM mechanisms are designed but unmodelled, so the alive-but-late failure is currently undetected until it becomes stale output. No ASIL is claimed, no HARA has been done, and every number here is a placeholder chosen on a dev bench – a binding FTTI comes from the hazard analysis, not from a demo. Stating that plainly is cheaper than having a reviewer find it.

§8.3 CROSSCUTTING – WIRE BUDGET

Why four events fit in one datagram each, and why the frame never became a fifth

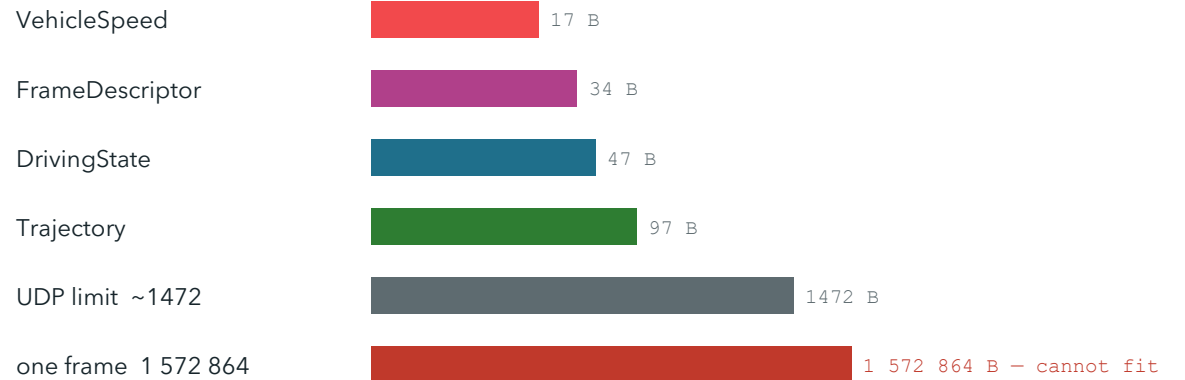
SOME/IP over UDP – the ceiling that matters

Ethernet MTU 1500 B → ~1472 B of UDP payload before IP fragmentation begins.

One lost fragment discards the whole datagram, and a fixed-rate stream has no retransmission – so a fragmented event is a silently dropped event.

Every modelled type here is sized to stay well inside that.

PAYLOAD, BYTES (log-ish scale, illustrative)



THE ARCHITECTURAL CONSEQUENCE AP R20-11 standardises SOME/IP and DDS network bindings in ARXML; a local zero-copy binding is implementation-specific. CAPI ships isoft/nsomeip and isoft/npc – no DDS. So the standardised option cannot carry a frame, and the option that could is not standardised. The frame ring is that gap made explicit rather than hidden: the interface is modelled, the buffer is not, and §11 records what that costs.

BOUNDED BY CONSTRUCTION

SteeringHorizon is a FIXED array of 20 floats – not variable-size. Nothing can grow the payload at run time.

E2E – STILL OPEN

isoft/e2e ships twelve profiles. A fixed-rate stream wants a counter and a timeout: P04 or P07. Not yet selected.

§9 ARCHITECTURE DECISIONS

Each with the alternative that was rejected – the part reviewers actually read

#	DECISION	ALTERNATIVE REJECTED	WHY	COST TO REVISIT
AD-1	Four AAs on the reference AD blocks	Two AAs – fuse capture into perception	Sensing is the sensor abstraction: radar and LiDAR arrive here later. Folding it into Perception hides that seam.	high
AD-2	Frames via shared memory + descriptor	Send the image as a SOME/IP event	1.5 MB against a ~1472 B datagram. Not a tuning problem.	low
AD-3	SOME/IP as the only binding	DDS, as AP R20-11 also standardises	CAPi ships no DDS binding. Fixed by what exists today, not by preference.	none
AD-4	Trajectory carries the whole 20-step horizon	Send the single next steering value	Control must keep steering after Planning is stopped. One value gives it nothing to decay.	low
AD-5	Two-phase startup by process argument	ExecutionDependency between processes	A dependency cannot span two function-group states. The state IS the barrier.	medium
AD-6	Control owns the chassis and the safe-stop	A separate safety supervisor process	A supervisor that is not the last thing before the actuator is not supervising anything.	high

AD-3 is not a preference – it is a consequence of what CAPi implements today. If a DDS binding appears, AD-2 is worth revisiting first.

Source: this project's AUTOSAR model — `av-stack/model/*.arxml`, validated against `AUTOSAR_00049.xsd` and generated with `ara-gen`

§10 QUALITY REQUIREMENTS · WHAT THE BENCH ACTUALLY SHOWED

Observed on WSL2 against a recorded clip – none of it is a target-hardware number

SCENARIO	STIMULUS → RESPONSE
Startup validation	Bad config path → init exits 1, names the file, AvPilotFG stays in Init
Service discovery	Four AAs started → all skeletons offered, all proxies found, < 1 s
Frame transport	instance=441054940 written by Sensing, mapped by Perception – same id
Nominal cadence	Planning ~10 Hz against Control 20 Hz → age cycles 0 → 51 → 104 ms
Planned degrade	SetState(Degraded) → planningd stops, three AAs keep running
Safe-stop on staleness	age > 200 ms → tyre 0.0000 rad, accel -4.0 m/s ² , sustained

STILL TO BE MEASURED

Capture → actuation, p50 / p99
SOME/IP latency under inference
Frame-drop rate (writer laps reader)
Memory headroom with daemons
All of the above on target, not WSL2

WHY NO TIMINGS ARE QUOTED

Debug SDK · CPU-only inference ·
no real-time kernel · shared host.
A latency figure from this bench
would be misleading, not cautious.

THE CONTROL EXPERIMENT THAT MATTERED

Upstream's single-process binary was built and run against the same config, same clip, same weights. That is what separated "the decomposition broke it" from "this configuration was never going to work" – and it settled a defect in minutes that guesswork had not settled in an hour. Keep it runnable.

§10.1 THE BENCH RUN

OpenLane clip test_open_lane_2 on WSL2 – the overlay at localhost:8080 and the planner telemetry behind it



tyre / accel = command pair · cte (raw) = cross-track error · epsi = heading error · kappa = curvature

```
[INFO] plan: tyre=-0.0000 rad accel=1.408 m/s² | cte=-0.13m(raw=-0.14m)
rad/s kappa=-0.0001 | cipo=true dist=108.4 m vel=+0.16 m/s
[INFO] plan: tyre=-0.0006 rad accel=1.414 m/s² | cte=-0.12m(raw=-0.10m)
rad/s kappa=-0.0002 | cipo=true dist=102.1 m vel=-0.66 m/s
[INFO] plan: tyre=-0.0006 rad accel=1.410 m/s² | cte=-0.11m(raw=-0.09m)
rad/s kappa=-0.0002 | cipo=true dist=97.1 m vel=-0.93 m/s
[INFO] plan: tyre=0.0014 rad accel=1.408 m/s² | cte=-0.10m(raw=-0.07m) c
ad/s kappa=0.0001 | cipo=true dist=97.0 m vel=-0.92 m/s
```

ON THE OVERLAY

green polygon = drivable path
red box + 113 m = the CIPO
cyan boxes = other detections

SERVED BY

upstream's own WebRTC page:
a debug HUD, outside the model

WHAT THIS CAPTURE SETTLES

Upstream's single-process binary,
same clip and weights – the
control experiment of §10.
The four AAs print the same line
with cte and epsi at exact zeros.

§11.1 RISKS & TECHNICAL DEBT

Open items, with the consequence of leaving each one open

OPEN ITEM	CONSEQUENCE IF LEFT	SEVERITY	NEXT STEP
H_resized gets the network size, not the capture size	Lane output projects as nonsense; cte and epsi read exact zeros, which looks like “perfectly centred” rather than an error	HIGH	add rawWidth/rawHeight to FrameDescriptor
PHM supervision unmodelled	The alive-but-late failure is undetected until it becomes stale output	HIGH	model a SupervisedEntity, wire the checkpoints
Init checks too weak	Perception's init validates nothing – it sits above the engine construction and reports ok regardless	MEDIUM	move the block below, check the assets
Control aborts at end of the speed source	A controller that throws when its speed feed stops is the failure it exists to prevent	MEDIUM	treat end-of-data as safe-stop
E2E profile not selected	No counter, no timeout, no CRC on any event	MEDIUM	P04 or P07 from isoft/e2e
Frame ring is outside the model	Half the data path is not analysable from the ARXML	LOW	revisit if a zero-copy binding appears
Perception has no ego-speed port	The debug HUD reads 0 mph by construction	LOW	leave it – a HUD is not worth a new dependency

Listed in the order they would be fixed. The first one is why the pipeline runs beautifully and steers on nothing.

Source: this project's AUTOSAR model — av-stack/model/*.arxml, validated against AUTOSAR_00049.xsd and generated with ara-gen

§11.2 WHAT THE DECOMPOSITION EXPOSED

Defects that existed in the monolith but could not surface until it was split

An implicit geometry coupling

Where Upstream had `frame.size()` in scope and passed it freely. Once Perception is a separate process it never sees the original frame – and the wrong size silently turns the projection matrix into the identity.

Roles `cte = 0.00 m` looks like success, not failure

A type collision the monolith hid

Where OpenCV declares `uint64 = uint64_t`; AUTOSAR's `Platform_Types.h` declares `unsigned long long`. Same width, different types, both global. No single-process build ever includes both.

Roles patched to the fixed-width spelling

Three rules for finding assets

Where `find_config` (config header), `find_config` (common/utils) and `find_model` search three different directories and throw the same message. Under EM the working directory is not the build tree.

Roles "Config file not found: X" names no directory

A duplicate symbol

Where `find_config` is declared with external linkage in one header and `DEFINED static` in another. Including both is an error – and no upstream translation unit does.

Roles `extern-then-static errors`; the reverse is legal

A missing link, upstream's too

Where `CppAD::local::temp_file` lives in `libcppad_lib`, which the planning module never links. Upstream's own binary does not link on this toolchain either.

Roles fixed for both targets in `patches/0002`

A loader rule worth knowing

Where `DT_RUNPATH` is not inherited by transitive dependencies and file capabilities make the loader ignore `LD_LIBRARY_PATH`. Both produce exit 127 with no application log at all.

Roles `-Wl,--disable-new-dtags`

THE POINT None of these was introduced by the port. Each was a latent assumption that one address space had been quietly satisfying. Splitting a process is, among other things, a way of finding out what your modules were really depending on – which is the argument for doing it before the hardware arrives, not after.

§12 GLOSSARY

Terms used in this deck

AA	Adaptive Application – a process built on ara:: APIs
ara::com	AP Communication Management: events, methods, fields
ARXML	AUTOSAR XML – the model exchange format
CAPI	AUTOSAR Common Adaptive Platform Implementation
EM / emd	Execution Management – starts and stops processes per state
SM / smd	State Management – owns function-group state transitions
PHM	Platform Health Management – alive and deadline supervision
Function Group	A named set of states; processes are listed per state
Executable	One binary; may back several Processes with different arguments

Process	A deployed instance of an Executable, with a StartupConfig
SWCL	Software Cluster – the deployable unit UCM would update
Skeleton / Proxy	Generated server / client side of a ServiceInterface
Event	One-way notification; latest sample matters, no get/set
InstanceSpecifier	Port name in the model, resolved to an instance at run time
seqlock	Lock-free scheme: odd generation = writing, even = stable
CIPO	Closest In-Path Object – the vehicle ahead in the ego lane
CTE	Cross-Track Error – lateral offset from the lane centre
MPC	Model Predictive Control – the planner's optimisation

THANK YOU

Nguyen Nguyen - baonguyenpro@gmail.com

`github.com/AUTOSAR/capi` · `github.com/autowarefoundation/vision_pilot` ·
`https://github.com/baonguyenhobby/Demo_VisionPilot_CAPI`

CAPi is released for information only; commercial exploitation requires an AUTOSAR licence.

